Analyse IV
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Introduction

Overview of the course

The course is a 4cts course taken by computer science, communication system, chemistry, environmental science. The course covers the following topics:

- Complex analysis, complex differentiation, complex integration, Laurent series, residue theorem
- Review of Fourier transform/series
- Laplace transform
- Applications to ODES/PDES (initial and Cauchy value problems)

Main textbook

The main book that this course refers to is '*Analyse avancée pour ingénieur*' by Dacorrogna Bernard and Tanteri Chiara.

Chapter 1

Review of Complex number $\mathbb{C}$

The goal of this chapter is to review what we have seen in Analysis I in order to be more comfortable when introducing new materials. As an addition I will add a bit of context on where and why complex analysis differs from multivariate analysis (this is not in the course but is always nice to have in order to better understand the materials).

Imaginary unit

As seen in previous classes, we define i the **imaginary unit** such that:

$$i^2 = -1$$

We also define $z \in \mathbb{C}$ as:

$$z = x + iy \qquad x, y \in \mathbb{R}$$

Furthermore:

$$x = \mathrm{Re} z \quad \text{real part}$$
$$y = \mathrm{Im} z \quad \text{imaginary part}$$

The field $\mathbb{C}$ can by fully mapped to $\mathbb{R}^2$, what this means is that every element in $\mathbb{C}$ can be directly '*translated*' to $\mathbb{R}^2$ **and** vis versa.

Complex Conjugate

Definition 1. Let us define the complex conjugate of z as:

$$\overline{z} = x - iy$$

Absolute Value

Definition 2. The absolute value (modulus) is defined as:

$$|z| = \sqrt{x^2 + y^2}$$

Intuition The reason for this definition really comes from Pythagoras theorem and the L_2 norm in $\mathbb{R}^2$.

Furthermore $|z_0 - z_1| = d(z_0, z_1)$ (the distance between those two number).
We can define domain in $\mathbb{C}$ as:

$$Q = \{z \in \mathbb{C} : |z - (1 + i)| < 1\}$$

Which is the disk centred in $z = 1 + i$ of radius 1.

Euler's Formula

Definition 3. For $\theta \in \mathbb{R}$:

$$e^{i\theta} = \cos\theta + i\sin\theta$$

So $e^{i\theta}$ lies on the unit circle around $z = 0$ (because $|e^{i\theta}| = 1$)

Definition 4. The Complex conjugate of the Euler formula is defined as

$$\begin{aligned}\overline{e^{i\theta}} &= \overline{\cos\theta + i \in \theta} \\ &= \cos\theta - i\sin\theta \\ &= \cos(-\theta) + i\sin(-\theta) \\ &= e^{-i\theta}\end{aligned}$$

Geometric intuition

What happens when doing the complex conjugate (if we see $\mathbb{C}$ as a plan in $\mathbb{R}^2$) is a symmetry around the x axis.

And this totally makes send when we recall on how it was defined when $z = x + iy$, $\overline{z} = x - iy$ we map the Im part to his opposite.

Remark 1. The rigorous definition of e^z is by the Taylor series. One should more see it as exp being **defined** by the Taylor serie and not the other way around. There is a great video about the definition of the exponential by 3Blue1Brown about taking the exponent of a matrix.

Algebraic Operations

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ we have the followings:

$$\begin{aligned}z_1 + z_2 &= (x_1 + iy_1) + (x_2 + iy_2) \\ &= (x_1 + x_2) + i(y_1 + y_2)\end{aligned}$$

$$z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2)$$

$$\begin{aligned}z_1 \cdot z_2 &= (x_1 + iy_1) \cdot (x_2 + iy_2) \\ &= x_1x_2 - y_1y_2 + i(y_1x_2 + x_1y_2)\end{aligned}$$

$$z \cdot \overline{z} = x^2 + y^2 = |z|^2 \in \mathbb{R}$$

$$\begin{aligned}\frac{z_1}{z_2} &= \frac{\overline{z_2}z_1}{\overline{z_2}z_2} \\ &= \frac{\overline{z_2}z_1}{|z_2|^2} \quad \text{if } z_2 \neq 0\end{aligned}$$

Polar coordinates

- If $z \in \mathbb{C}$ then $z = |z|e^{i\theta}$ for some $\theta \in \mathbb{R}$.
- This also means that $z^k = |z|^k e^{ik\theta}$ for $k \in \mathbb{Z}$

Let us define now our two complex numbers in polar coordinates $z_1 = |z_1|e^{i\theta_1}, z_2 = |z_2|e^{i\theta_2}$. If we want to apply algebraic operations defined previously to our new notation we would

have:

$$z_1 \cdot z_2 = |z_1||z_2|e^{i(\theta_1+\theta_2)}$$

Dividing our two numbers really comes down to multiply by the inverse which is defined as $z_2^{-1} = \frac{1}{|z_2|}e^{-i\theta_2}$
Lastly:

$$z_1 = z_2 \iff |z_1| = |z_2| \wedge \theta_1 - \theta_2 = 2\pi \cdot k, \quad k \in \mathbb{Z}$$

Geometric Intuition Multiplication by $e^{i\theta_0}$ corresponds to a rotation by an angle θ_0.

Remark 2. Therefore as we can see any number in $\mathbb{C}$ requires only two variables, his norms and argument, which means that when wanting to express any complex number, one only need to have the angle (the rotation) and the modulus. This can be very interesting when our goal is to visualize complex functions. In order to have a good visualization, we need 4 dimensions: two for the input, two for the output. But if all we need is the angle $\theta \in [-\pi, \pi]$ and $|z| \in \mathbb{R}_+$ then we can map using colors for the angle and the third axis for the modulus. This way we can directly see where our number collaps to 0 or when it goes to $\pm\infty$. (there is a great youtube video about 5 ways to visualize complex functions that I recommend).

Principal argument

Definition 5. Let $z = |z|e^{i\theta} \in \mathbb{C}$, the principal argument is $]\theta' \in -\pi, \pi]$ such that:

$$z = |z|e^{i\theta'}$$

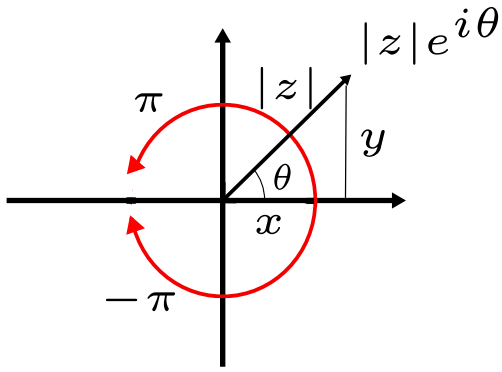


Figure 1.1: Polar representation

Example

Let us try to solve the following equation:

$$z^5 = 2$$

To do so let us write our number z into polar coordinate: $z = |z|e^{i\theta}$. If we insert our number into our equation we have:

$$2 = |z|^5 e^{i5\theta}$$

As $|e^{i5\theta}| = 1$ we have that

$$2 = |z|^5 \implies |z| = 2^{\frac{1}{5}}$$

And as 2 is a real number this implies that $5\theta = 2\pi k \implies \theta = \frac{2\pi k}{5}$ where $k \in \mathbb{Z}$ and if we want to take the principal arguments, we have as solution:

$$\theta \in \{-\frac{4\pi}{5}, -\frac{2\pi}{5}, 0, \frac{2\pi}{5}, \frac{4\pi}{5}\}$$

More generally, a k-th order equation will gives us k solutions.

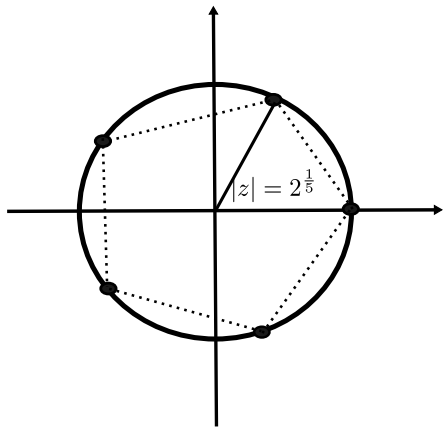


Figure 1.2: Solution of $z^5 = 2$

1.1 Construction of $\mathbb{C}$ (not in the course)

The goal of this section is to add a bit more context on the $\mathbb{C}$ field. First let us recall how real analysis worked. We used the field $(\mathbb{R}, +, \cdot)$ which is an **ordered** field, i.e. we have an ordering on our elements. Formally:

Definition 6. Ordered field

$\forall x, y \in \mathbb{R}$ precisely one of $x < y, x > y, x = y$ holds true.

- $x, y > 0 \implies x \cdot y > 0$

This is the axiomatic definition of an ordered field.

And as we already know, $\mathbb{C}$ is not an ordered field, $2 + i$ cannot be greater nor smaller to 12.

One of the reason on why $\mathbb{C}$ has come up is that the field $(\mathbb{R}, +, \cdot)$ is **not algebraically closed**. We can find polynomials with real coefficient which don't have real roots. $x^2 + 1$ has not root over $\mathbb{R}$. This is a major issue, we need to find a field which is algebraically closed. The question now that was asked: does $\left(\mathbb{R}^2, +\right)$ (which is already an abelian group) can be turned into a field $\left(\mathbb{R}^2, +, ?\right)$? And what about $(\mathbb{R}^n, +)$ is there a way to 'create' a multiplication which would be suitable?

For $n = 1$ this is just the real number. For $n = 2$ the resulting field you will get is $\mathbb{C}$. For $n = 4$ the answer is the Hamilton group (kind of) and the same for $n = 8$. For all other number it is just **impossible**! And this is pretty bizarre, it worked for $1, 2, 4, 8$ and then it stopped.

Theorem 1. If we can create a field on $\mathbb{R}^2$ then

$$\exists z \in \left(\mathbb{R}^2, +, \cdot\right) \text{ s.t. } z^2 = -1$$

Proof. Let us choose a basis element $\{1, e\}$ of $\mathbb{R}^2$. This directly implies that:

$$z = x \cdot 1 + y \cdot e \text{ , for some } x, y \in \mathbb{R}$$

If we want to multiply our number by itself:

$$z^2 = x^2 \cdot 1 + 2xy \cdot e + y^2 \cdot e^2$$

Remark 3. A strange thing we did here is to add the $2xy \cdot e$. Recall that previously, when we had a vector in $\mathbb{R}^2, x$ that we squared, the output that we will get would be:

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1^2 + x_2^2$$

Or if we have written them using the basis vector $e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ this would gives us:

$$(x_1 e_1 + x_2 e_2) \cdot (x_1 e_1 + x_2 e_2)$$

The important thing is that both equation above are **totally equivalent**. If we develop this we will get:

$$x_1^2 e_1^2 + 2x_1 x_2 e_1 e_2 + x_2^2 e_2^2$$

And because e_1 and e_2 are orthogonal, by definition $e_1 e_2 = 0$:

$$= x_1^2 e_1^2 + x_2^2 e_2^2 = x_1^2 + x_2^2$$

Now we want to rewrite e^2 using the basis that we introduced before:

$$e^2 = a \cdot 1 + b \cdot e, \qquad a, b \in \mathbb{R}$$

This implies that:

$$z^2 = \left(x^2 + ay^2\right) \cdot 1 + \left(2xy + by^2\right) \cdot e$$

Now we want to show that there exists the pair $(x, y) \in \mathbb{R}^2$ such that $z^2 = -1$. Then we must have the following:

$$2xy + by^2 = 0 \iff \begin{cases} y = 0 \\ x = -\frac{b}{2} y \end{cases} \qquad \text{or}$$

But here $y = 0$ is not sufficient because this implies for the second relation:

$$x^2 + ay^2 = -1$$

that $x^2 = -1$ where $x \in \mathbb{R}$ which is impossible. We therefore look into the second case, $y \neq 0$ and $x = -\frac{b}{2} y$. In this case this implies that:

$$\begin{aligned} z^2 &= \left(-\frac{b}{2} y\right)^2 + ay^2 \cdot 1 + \underbrace{\left(2\left(\frac{-b}{2}\right) y^2 + by^2\right) \cdot e}_{=0} \\ &= \left(a + \frac{b^2}{4}\right) y^2 \cdot 1 \end{aligned}$$

The second claim is that $a + \frac{b^2}{4} < 0$

Proof. The proof will be done by contradiction. Suppose that $a + \frac{b^2}{4} \geq 0$. This implies that we can take the square root (in the sense of $\mathbb{R}$) and therefore we have that:

$$z = \pm y\sqrt{a + \frac{b^2}{4}}$$

Which implies that:

$$\left(z - y\sqrt{a + \frac{b^2}{4}}\right)\left(z + y\sqrt{a + \frac{b^2}{4}}\right) = 0$$

This implies that either $z = y\sqrt{a + \frac{b^2}{4}} = x \cdot 1 + y \cdot e$ or the same with a negative sign.

$$\pm y\sqrt{a + \frac{b^2}{4}} \cdot 1 = x \cdot 1 + y \cdot e$$

Since $\{1, e\}$ are linearly independent, this equality is impossible unless $y = 0$ which contradicts the case we are studying ($y \neq 0$). Therefore $a + \frac{b^2}{4} < 0$ □

Now our goal is to proof there exists a y such that $\left(a + \frac{b^2}{4}\right)y^2 = -1$. We can set

$$y := \frac{1}{-\sqrt{a + \frac{b^2}{4}}}, \qquad x = \frac{-b}{2}y$$

This directly implies that:

$$\begin{aligned} z^2 &= \left(a + \frac{b^2}{4}y^2\right) \\ &= \left(a + \frac{b^2}{4}\right)\left(\frac{-1}{a + \frac{b^2}{4}}\right) = -1 \end{aligned}$$

□

The claim we just prove is not trivial at all, if we want to turn $\mathbb{R}^2$ into a field, then there has to be an element of the field such that $d = -1$. This also implies that our multiplication will be a new one which will be pretty counter-intuitive for us. Now let us pick our element $z \in \left(\mathbb{R}^2, +, \cdot\right)$ such that $z^2 = -1$, and let us call it $\mathbf{i}$.

Now we can define our field $\mathbb{C}$:

Definition 7. Let us define the **complex field** $\mathbb{C}$:

$$\begin{aligned} \mathbb{C} &:= \mathbb{R}[i] := \{a + bi, \quad a, b \in \mathbb{R}\} \\ &= \mathbb{R}[X]/(X^2 + 1) := \{a + bX \in \mathbb{R}[X]\} \end{aligned}$$

$\mathbb{C}$ is algebraically closed

Theorem 2. Any $P \in \mathbb{C}[X]$ non-zero (not the zero polynomial) has at least a root in $\mathbb{C}$.

Proof. To prove this we will do it by contradiction. Assume that P is a non-constant polynomial and suppose it **has no root** in $\mathbb{C}$. Then this implies that:

$$P(z) \neq 0 \quad \forall z \in \mathbb{C}$$

Therefore, we can define:

$$f(z) = \frac{1}{P(z)}$$

Since $P(z) \neq 0$ everywhere and P is holomorphic, f is also holomorphic on all of $\mathbb{C}$ (entire).

The next step of our proof is to show that f is bounded. Let

$$P(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0$$

For large $\|z\|$, the leading term dominates:

$$\|P(z)\| \approx \|a_n\| \|z\|^n$$

More precisely, there exists R such that if $\|z\| > R$ then:

$$\|P(z)\| \geq \frac{\|a_n\|}{2} \|z\|^n$$

Thus:

$$\|f(z)\| = \frac{1}{\|P(z)\|} \leq \frac{2}{\|a_n\| \|z\|^n}$$

Therefore, when $\|z\| \to \infty$ we have that

$$f(z) \to 0$$

Hence f is bounded outside a large disk. And inside $\|z\| < R$, the continuous function f on a compact set is also **bounded**. Therefore, f is bounded on all of $\mathbb{C}$.

All the work is done now, all we have to do is to apply Liouville's theorem:

Theorem 3. Every bounded entire function is constant

Which therefore implies that f must be constant. But we just said

$$f(z) = \frac{1}{P(z)}$$

which means that P is constant which is a contradiction.

Conclusion:

Every non-constant polynomial in $\mathbb{C}[x]$ has a complex root.

$\square$

Chapter 2

Complex Functions

Definition 8. We call f a complex function as:

$$\begin{aligned} f : \mathbb{C} &\to \mathbb{C} \\ z \mapsto f(z) &= f(x+iy) \\ &= u(x,y) + iv(x,y) \end{aligned}$$

In particular we have $u, v : \mathbb{R}^2 \to \mathbb{R}$

A nice thing to see here is the relation between $\mathbb{R}^2$ and $\mathbb{C}$. As said before we can always map a complex number to a number in $\mathbb{R}^2$. By 'splitting' our function into to two sub functions u, v which map to $\mathbb{R}$. We effectively transformed our function $f : \mathbb{C} \to \mathbb{C}$ into two sub function that goes from $\mathbb{R}^2 \to \mathbb{R}$ which therefore point to $\mathbb{R}^2$ (if we replaced the 1 by a unit vector, and i as another unit vector).

Examples

First let us see the simplest example, the **unit** function: $f(z) = z = x + iy$. In this case we can easily see that

$$u(x,y) = x \qquad v(x,y) = y$$

A bit more sophisticated function could be for instance: $f(z) = \overline{z} = x - iy$. Which gives us:

$$u(x,y) = x \qquad v(x,y) = -y$$

If we take the square we have $f(z) = z^2 = (x+iy)^2 = \left(x^2 - y^2\right) + i2xy$

In all of those expression we can easily see where our function u and v are. But is it always the case?

Trigonometric Functions

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}$$

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

Remark 4. We have already seen the same concept in Analysis III. We use a lot those definition when dealing with Fourier series. (with odd/even series)

$$\cosh z = \frac{e^z + e^{-z}}{2}$$

$$\sinh z = \frac{e^z - e^{-z}}{2}$$

The rigorous definition of e^z is by the Taylor series. One should more see it as exp being **defined** by the Taylor serie and not the other way around. There is a great video about the definition of the exponential by 3Blue1Brown.

2.1 Limit and Continuity

Definition 9. We say $f : \mathbb{C} \to \mathbb{C}$ is **continuous** at a point $z_0 \in \mathbb{C}$ if $\forall \varepsilon > 0$ there exists $\delta > 0$ such that $\forall z \in \mathbb{C}$ with $\|z - z_0\| \leq \delta$ we have:

$$\|f(z) - f(z_0)\| \leq \varepsilon$$

Equivalently:

$$\lim_{z \to z_0} f(z) = f(z_0)$$

Remark 5. This definition is directly mapped to the one seen in Analysis II. What we are really doing here in fact is the limit of our function the same way as we would have done in $\mathbb{R}^2$. If we develop this:
Let $f(z) = f(x + iy) = u(x, y) + iv(x, y)$ then f is continuous at $z_0 = x_0 + y_0 i$ if and only if u and v are continuous at (x_0, y_0)

Operation on Continuous Functions

If f, g are continuous functions then the same is true for:

- $f + g$
- $f \cdot g$
- $\overline{f}$
- $f \circ g$
- $\vdots$

2.1.1 Complex Derivative

Definition 10. We say $f : \mathbb{C} \to \mathbb{C}$ is **complex differentiable** at a point $z_0 \in \mathbb{C}$ if:

$$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} \text{ exists}$$

We denote this limit by $f'(z_0)$

Definition 11. If f is differentiable at every point of an open set $\omega \subseteq \mathbb{C}$, we say f is **holomorphic** on ω.

Definition 12. If f is holomorphic on $\mathbb{C}$, we call f **entire**.

Remark 6. The same rules of differentiation of products (product rule, Leibniz rule), Compositions (chain rule), etc. are as in the real case.

Recall on topology Open set

Definition 13. $\omega \subseteq \mathbb{C}$ is **open** if for all $z_0 \in \omega$, there exists $\Gamma > 0$ such that the disk $D_\Gamma(z_0) = \{z \in \mathbb{C} : \|z - z_0\| < r\} \subseteq \omega$

Examples

- $\mathbb{C}$ is open
- $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$ is open
- $\{z \in \mathbb{C} : \text{Re } z \geq 0\}$ is not open

Examples of Differentiation

Let $f(z) = z$ our goal is to compute the derivative of f for all $z_0 \in \mathbb{C}$. By the definition:

$$f'(z_0) = \lim_{z \to z_0} \frac{z - z_0}{z - z_0} = 1$$

In particular, f is (holomorphic and) entire.

Let us try a little more sophisticated example. Let $f(z) = \frac{1}{z}$. For all $z_0 \in \mathbb{C}^*$:

$$f'(z_0) = -\frac{1}{z_0^2}$$

So f is holomorphic on $\mathbb{C}^*$ but not entire.

2.2 Cauchy-Riemann equations

The goal of those equations is to check if a function f is differentiable or/and is holomorphic. When we have quite 'easy' function (as seen before) showing that the function is holomorphic by taking the limit is possible. However when function become tedious we will use Cauchy-Riemann questions:

Definition 14. Let $w \subseteq \mathbb{C}$ open, $f : w \to \mathbb{C}$ let:

$$z = x + iy \mapsto f(z) = u(x, y) + v(x, y)i$$

Be holomorphic. Then, $u, v \in C^1(w)$ (the space of differentiable function $w \subseteq \mathbb{R}^2 \to \mathbb{R}$) with continuous derivative) and:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$
$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Those equation are called Cauchy-Riemann equations.

Theorem 4. If the relation:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$
$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

holds then f is holomorphic

Proof of the direct implication ' $\Longrightarrow$ '

Suppose that f is holomorphic on w, therefore, $\forall z_0 \in w$ we have:

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} \quad \text{exists}$$

We will calculate the limit in two ways.

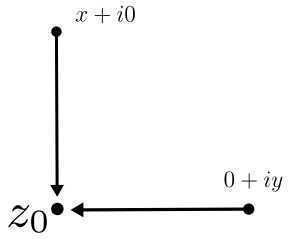


Figure 2.1: Two ways of approaching

First we will approaching z_0 from the x-direction:

$$\begin{aligned} f'(z_0) &= \lim_{x \to x_0} \frac{f(x + y_0 i) - f(x_0 + y_0 i)}{x + y_0 i - (x_0 + y_0 i)} \\ &= \lim_{x \to x_0} \frac{u(x, y_0) + iv(x, y_o) - u(x_0, y_0) - iv(x_0, y_0)}{x - x_0} \\ &= \underbrace{\lim_{x \to x_0} \frac{u(x, y_0) - u(x_0, y_0)}{x - x_0}}_{= \frac{\partial u}{\partial x}(x_0, y_0)} + \underbrace{\lim_{x \to x_0} \frac{v(x, y_0) - v(x_0, y_0)}{x - x_0} i}_{= \frac{\partial v}{\partial x}(x_0, y_0)} \end{aligned}$$

Now we will approach z_0 from the $y-$direction:

$$\begin{aligned} f'(z_0) &= \lim_{y \to y_0} \frac{f(x_0, yi) - f(x_0 + y_0 i)}{x_0 + yi - (x_0 + y_0 i)} \\ &= \lim_{y \to y_0} \frac{u(x_0, y) + iv(x_0, y) - u(x_0, y_0) - iv(x_0, y_0)}{i(y - y_0)} \\ &= \overbrace{\frac{1}{i}}^{=-i} \underbrace{\lim_{y \to y_0} \frac{u(x_0, y) - u(x_0, y_0)}{y - y_0}}_{= \frac{\partial u}{\partial y}(x_0, y_0)} + \frac{i}{i} \underbrace{\lim_{y \to y_0} \frac{v(x_0, y) - v(x_0, y_0)}{y - y_0}}_{= \frac{\partial v}{\partial y}(x_0, y_0)} \end{aligned}$$

By comparing real and imaginary part leads to the Cauchy-Riemann equations. Moreover form this proof, we get explicit expression for $f'(z_0)$:

$$f' = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial x} i = \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} i$$

Example

$f : \mathbb{C}^* \to \mathbb{C},\ f(z) = \frac{1}{z}$

$$\begin{aligned} f(z) &= \frac{\overline{z}}{\|z\|^2} \\ &= \frac{x}{x^2+y^2} - \frac{y}{x^2+y^2} i \\ &= u(x,y) + v(x,y)i \end{aligned}$$

Now we will compute the Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = \frac{x^2+y^2-2x^2}{\left(x^2+y^2\right)^2} \qquad \frac{\partial v}{\partial x} = \frac{2xy}{\left(x^2+y^2\right)^2}$$

$$\frac{\partial u}{\partial y} = -\frac{2xy}{\left(x^2+y^2\right)^2} \qquad \frac{\partial v}{\partial y} = \frac{-\left(x^2+y^2\right)+2y^2}{\left(x^2+y^2\right)^2}$$

We therefore the Cauchy-Riemann equations satisfied and:

$$\begin{aligned} f' &= \frac{\partial u}{\partial x} + \frac{\partial v}{\partial x} i \\ &= \frac{y^2-x^2}{\left(x^2+y^2\right)^2} + \frac{2xy}{\left(x^2+y^2\right)^2} i \end{aligned}$$

Indeed, this coincides with $-\frac{1}{z^2}$ (computation).

Example

The last example will be: $f(z) = e^z$ let us massage our function:

$$\begin{aligned} f(z) &= e^{x+yi} = e^x e^{yi} \\ &= e^x \cos y + e^x \sin(y) i \\ &= u(x,y) + iv(x,y) \end{aligned}$$

Not let us compute the Cauchy-Riemann equations:

$$\begin{aligned} \frac{\partial u}{\partial x} &= e^x \cos y \qquad & \frac{\partial v}{\partial x} &= e^x \sin y \\ \frac{\partial u}{\partial y} &= -e^x \sin y \qquad & \frac{\partial v}{\partial y} &= e^x \cos y \end{aligned}$$

As we can see that equation holds which implies that our function f is holomorphic.

Moreover, the derivative of our function is given as:

$$f' = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = e^x \cos y + ie^x \sin y = e^z$$

2.2.1 Rules for differentiation

Theorem 5. Let $f, g : w \to \mathbb{C}$ holomorphic. Then:

$$\begin{aligned} (f+g)' &= f' + g' \\ (fg)' &= f'g + fg' && \text{Leibniz rule} \\ \left(\frac{f}{g}\right)' &= \frac{f'g - fg'}{g^2} && g \neq 0 \end{aligned}$$

If the image of g is inside of w:

$$(f \circ g)'(z) = f'(g(z)) \cdot g'(z) \qquad \text{chain rule}$$

All of these relations can be shown exactly as in the real case, we will do $(fg)'(z_0)$. First let us use the definition of the derivative:

$$\begin{aligned} (fg)'(z_0) &= \lim_{z \to z_0} \frac{f(z)g(z) - f(z_0)g(z_0)}{z - z_0} \\ &= \lim_{z \to z_0} \frac{f(z)g(z) - f(z_0)g(z) + f(z_0)g(z) - f(z_0)g(z_0)}{z - z_0} \\ &= \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} g(z) + f(z_0) \frac{g(z) - g(z_0)}{z - z_0} \\ &= f'(z_0)g(z_0) + f(z_0)g'(z_0) \end{aligned}$$

With these rules and previous examples, it follows the usual formulas for differentiation of polynomial, trigonometric functions, etc...

$$\begin{aligned} (z^k)' &= kz^{k-1} \\ (\cos z)' &= -\sin z \\ (\sin z)' &= \cos z \end{aligned}$$

Geometric interpretation of Cauchy-Riemann equations

Let's fix $f : \mathbb{C} \to \mathbb{C}$ holomorphic, and view it as a function $f : \mathbb{R}^2 \to \mathbb{R}^2$. Geometrically the Jacobian matrix:

$$Df = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} \end{pmatrix}$$

tells us how f acts on 'infinitesimal rectangles'/ disturb volume.

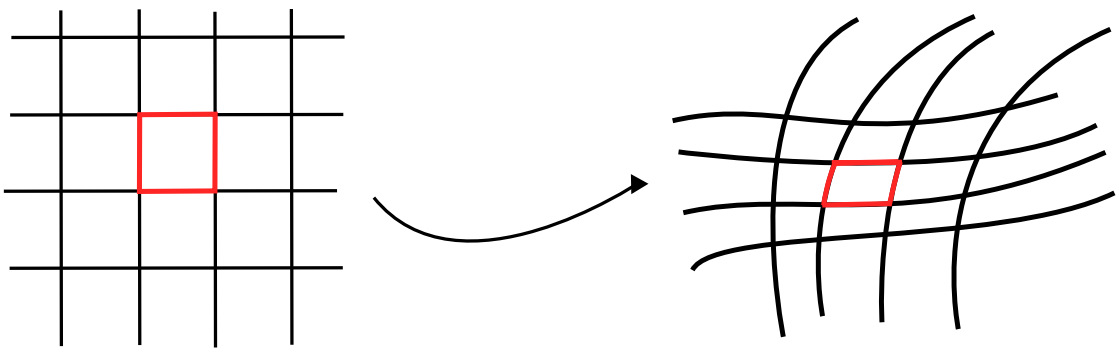


Figure 2.2: Transformation of the Jacobian matrix

A very specific deformation happens for holomorphic functions. In fact, the Cauchy-Riemann equations imply that the Jacobian is in the form of:

$$\begin{aligned} Df &= \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \\ &= \sqrt{a^2+b^2} \underbrace{\begin{pmatrix} \frac{a}{\sqrt{a^2+b^2}} & -\frac{b}{\sqrt{a^2+b^2}} \\ \frac{b}{\sqrt{a^2+b^2}} & \frac{a}{\sqrt{a^2+b^2}} \end{pmatrix}}_{\text{rotation matrix}} \end{aligned}$$

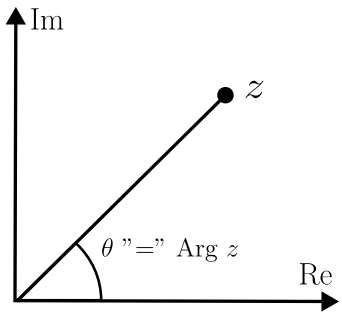
Recall that a rotation matrix is in the form of:

$$\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

Hence we have that $Dg = \lambda\cdot$rotation! Therefore, infinitesimal square is mapped to a square.

2.2.2 Complex logarithm

First let us recall what the principal argument is. For $z \neq 0$ we have that $-\pi < \text{Arg } z \leq \pi$



It can be computed by:

$$\text{Arg}z = \begin{cases} \arctan\frac{y}{x} & \text{if } x > 0 \\ \frac{\pi}{2} & \text{if } x = 0, y > 0 \\ \pi + \arctan\frac{y}{x} & \text{if } x < 0, y \geq 0 \\ -\frac{\pi}{2} & \text{if } x = 0, y < 0 \\ -\pi + \arctan\frac{y}{x} & \text{if } x < 0, y < 0 \end{cases}$$

Arg is (defined yet) **discontinuous** on the negative real axis, it jumps from π to $-\pi$ across the negative real axis.

Okay this is nice but why are we even talking about this here in the first case, we are trying to define the logarithm for complex number. How de we define complex logarithm? First what does it do? It is the inverse of the exponential meaning that we should have $\log e^z = z$. If we expand z into Cartesian form, $\log\left(e^{x+iy}\right) = x + iy$.

If $\log z = x + iy$ then we should have $z = e^{x+iy} = e^x e^{yi}$. By polar coordinate, we can also define z as:

$$z = \underbrace{\|z\|}_{=e^x} \underbrace{e^{i\text{Arg}z}}_{=e^{iy}}$$

Remark 7. We are allowed to do so because $z = e^{x+iy}$. In general we don't have $y =$Arg z. Now we have something that could be put into the log:

Definition 15. We define:

$$lgz = lg\|z\| + i\text{Arg}z$$

Complex logarithm, more precisely the principal branch of complex algorithm

Remark 8.
- $\log z$ is not defined at $z = 0$
- $\log z$ is discontinuous on the negative real axis

More precisely, computing $\log z$ for $z = x + iy$ with $x > 0$:

$$\begin{aligned}\log z &= \log\sqrt{x^2+y^2} + i\arctan\frac{y}{x}\\ &= u(x,y) + iu(x,y)\end{aligned}$$

It is easy to verify that log is holomorphic on $\mathbb{C} \setminus \{Im = 0, Re \leq 0\}$ and $(\log z)' = \frac{1}{z}$

logarithm rules

$$\begin{aligned}\log(z_1 z_2) &= \log(z_1) + \log(z_2)\\ \log(z^\gamma) &= \gamma\log(z) \qquad \gamma \in \mathbb{C}\end{aligned}$$

Example

If $z_1 = z_2 = -1$ then $\log z_1 = \log z_2 = i\pi$. (since $e^{i\pi} = -1$). Yet we have that $\log(z_1 z_2) = 0 \neq \log(z_1) + \log(z_2)$. Similarly

$$\log\left(z_1^2\right) \neq 2\log(z_1)$$

However these rules remain true up to $2\pi i\mathbb{Z}$:

$$\begin{aligned}\log(z_1 z_2) - \log z_1 - \log z_2 &\in 2\pi i\mathbb{Z}\\ \log(z^\gamma) - \gamma\log(z) &\in 2\pi i\mathbb{Z}\end{aligned}$$

2.3 Complex integration

Let's first recall from vector analysis: if $\vec{F} : \mathbb{R}^2 \to \mathbb{R}^2$ is a vector field such that

$$F(x_1, x_2) = (F_1(x_1, x_2), F_2(x_1, x_2))$$

And $\gamma : [a, b] \to \mathbb{R}^2$ is a C^1 curve with image Γ then:

$$\underbrace{\int_\Gamma F dl}_{\text{line integral}} := \int_a^b F(\gamma(t)) \cdot \gamma'(t) dt$$

Here the orientation is **important**. We will need to recall some facts:

1. The line integral is **independent** of the parametrisation, but switching orientation creates a negative sign.
2. If $\gamma : [c, d] \to [a, b]$ is one-to-one, out and increasing, then γ is a parametrization of Γ.

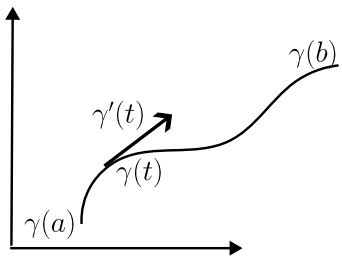


Figure 2.3: Sorry for the curve

3. If γ is closed, meaning $\gamma(a) = \gamma(b)$ and $\text{curl}F = 0$ $(= \nabla \times F)$ in the 'interior' of Γ, then we have that:

$$\int_\Gamma F dl = 0$$

   This is a consequence from Stokes theorem
4. A curve is called:
   - **simple** if it has no self intersections
   - **regular** if $\gamma' \neq 0$ $\forall x \in [a, b]$

Now we will define a similar notion of line integrals for complex functions.

Definition 16. Let $\gamma : [a, b] \to \mathbb{C}$ be a regular curve that parametrises $\Gamma \subseteq \mathbb{C}$. Let $f : \Gamma \to \mathbb{C}$ be continuous.

Then we define the line integral of f along Γ by:

$$\int_\Gamma f(z)dz = \int_\gamma f(z)dz := \int_a^b f(\gamma(t)) \cdot \gamma'(t)dt$$

Remark 9. The $\int_\gamma$ is a slight abuse of notation

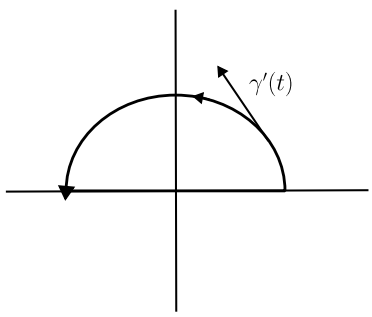
Usually we write $-\Gamma$ to mean the curve Γ with the **reverse** orientation. In this case we have:

$$\int_{-\Gamma} f(z)dz = -\int_\Gamma f(z)dz$$

Remark 10. The value of this line integral is independent of the parametrization of γ as long as the orientation is preserved.

Example

Let Γ be the unit half circle around $z = 0$, parametrized counter clockwise:



$$\begin{aligned}\gamma : [0, \pi] &\to \mathbb{C} \\ \gamma(t) &= e^{it} \\ \gamma'(t) &= ie^{it}\end{aligned}$$

We seek to integrate $f(z) = z^2$ over Γ:

$$\begin{aligned}\int_\Gamma z^2 dz &= \int_0^\pi \left(e^{it}\right)^2 ie^{it} dt \\ &= i\int_0^\pi e^{3it} dt = i\left[\frac{e^{3it}}{ti}\right]_0^\pi = -\frac{2}{3}\end{aligned}$$

Remark Above we used the fundamental theorem of calculus. For more general function, one splits the integrands into real and imaginary part and computes the two real integrals separately.

Example ii

Let Γ be the unit circle oriented counter-clockwise. We will use the same γ as before but we will extend its domain of definition:

$$\begin{aligned}\gamma : [0, 2\pi] &\to \mathbb{C} \\ \gamma(t) &:= e^{it}\end{aligned}$$

With again $\gamma'(t) = ie^{it}$. We try to compute $f(z) = z^2$:

$$\begin{aligned}\int_\Gamma z^2 dz &= \int_0^{2\pi} \left(e^{it}\right)^2 ie^{it} dt \\ &= \int_0^{2\pi} e^{3it} dt \\ &= i\left[\frac{e^{3it}}{3i}\right]_0^{2\pi} = 0\end{aligned}$$

This is a special case of the Cauchy integral theorem (that we will see later).

Example iii

Let Γ and γ be defined as above. We will take this time $f(z) = \frac{1}{z}$.

$$\begin{aligned}\int_\Gamma \frac{1}{z} dz &= \int_0^{2\pi} \frac{1}{e^{it}} ie^{it} dt \\ &= \int_0^{2\pi} i dt = 2\pi i \neq 0\end{aligned}$$

Example iv

Now let $f(z) = \frac{1}{z^2}$:

$$\begin{aligned}\int_\Gamma \frac{1}{z^2} &= \int_0^{2\pi} \frac{1}{\left(e^{it}\right)^2} ie^{it} dt = i\int_0^{2\pi} e^{-it} dt \\ &= -\left[e^{it}\right]_0^{2\pi} = 0\end{aligned}$$

2.3.1 Cauchy's Theorem

The Cauchy theorem tells us what the value of a complex integral is on general constraints:

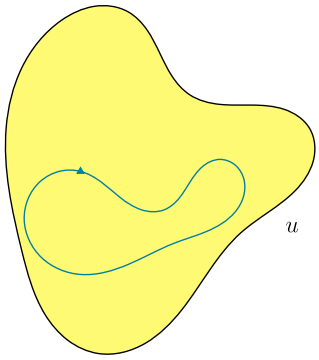
Theorem 6. Let $U \subseteq \mathbb{C}$ open and simply connected set and let Γ be a closed and piecewise regular curve inside U. Let $f : U \to \mathbb{C}$ be holomorphic. Then:

$$\int_\Gamma f(z)dz = 0$$

Geometric intuition

Definition 17. A domain is said simply connected when it is a domain with only one component and '*no holes*' e.g. $\mathbb{C}, \{z \in \mathbb{C} : \|<\| \leq 1\}$, while $\mathbb{C}^*$ is not simply connected.

Definition 18. **Piecewise regular** Γ means Γ might have finitely many curves, e.g. square



Why is this true?

Let $\gamma(t) = \alpha(t) + \beta(t)i$ a parametrization Γ_1 and:

$$f(x + yi) = u(x, y) + iv(x, y)$$

Then the integral gives us:

$$\begin{aligned}
\int_\Gamma f(z)dz &= \int_a^b f(\gamma(t)) \cdot \gamma'(t)dt \\
&= \int_a^b [u(\gamma(t)) + iv(\gamma(t))] \cdot [\alpha'(t) + \beta'(t)i]dt \\
&= \int_a^b [u(\gamma(t))\alpha'(t) - v(\gamma(t)) - \beta'(t)]dt \\
&+ i \int_a^b [u(\gamma(t))\beta'(t) + v(\gamma(t))\alpha'(t)]dt \\
&= \int_a^b \begin{pmatrix} u(\gamma(t)) \\ -v(\gamma(t)) \end{pmatrix} \cdot \begin{pmatrix} \alpha'(t) \\ \beta'(t) \end{pmatrix} dt + i \int_a^b \begin{pmatrix} v(\gamma(t)) \\ u(\gamma(t)) \end{pmatrix} \cdot \begin{pmatrix} \alpha'(t) \\ \beta'(t) \end{pmatrix} dt \\
&= \int_a^b F dl + i \int_a^b G dl
\end{aligned}$$

And by Green's theorem we have:

$$= \int_{iut\Gamma} \text{curl} F dxdy + i \int_{iut\Gamma} \text{curl} G dxdy$$

And if we compute the curl:

$$\text{curl}F = \frac{\partial F_2}{\partial x_1} - \frac{\partial F_1}{\partial x_2} = -\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$$
$$\text{curl}F = \frac{\partial G_2}{\partial x_1} - \frac{\partial G_1}{\partial x_2} = \frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}$$

And by the Cauchy Riemann equations both are equal to 0. Hence:

$$\int_\Gamma f(z)dz = 0$$

Notice here we used that since u is simply connected and $\Gamma \subseteq u$ we have int $\Gamma \subseteq u$, so F, G are defined on int Γ

Example

Let Γ be the unit circle, $f(z) = z^2$. By applying Cauchy theorem to $u = \mathbb{C}$, then:

$$\int_\Gamma f(z)dz = 0$$

Example ii

Again, let's take Γ to be the unit circle and $f(z) = \frac{1}{z}$.

We are looking for a simply connected domain which contains the unit circle **and** f is holomorphic. But this is not possible because f is only holomorphic on $\mathbb{C}^*$ and there is no simply connected domain $U \subseteq \mathbb{C}^*$ that contains Γ. Therefore, Cauchy's theorem does not apply. (indeed, as we have calculated above, $\int_\Gamma f(z)dz \neq 0$)

Example iii

Let $f(z) = \frac{1}{z}$, $\Gamma = \{z \in \mathbb{C} : \|z - 2\| = 1\}$.

For instance let us take the domain $u = \{\text{Re} > 0\}$ which is a simply connected domain which contains Γ, and where f is holomorphic on u. By the Cauchy theorem, we have that:

$$\int_\Gamma f(z)dz = 0$$

In practice, we'll apply Cauchy's theorem as follows:

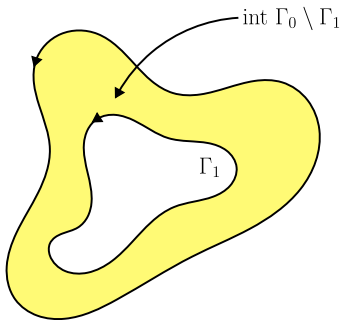
- Given Γ and f
- Find a simply connected open domain $U \subseteq \mathbb{C}$ such that:
  - U contains Γ and
  - f is holomorphic on U

There is an extension of Cauchy's theorem which is nice for application:

Theorem 7. Let Γ_0, Γ_1 be closed and simple regular curves such that $\Gamma_1 \subseteq$ int Γ_0. Let γ_0, γ_1 be parametrization of Γ_0, Γ_1 with the same orientation.

If f is holomorphic on our two curves, then:

$$\int_{\Gamma_0} f(z)dz = \int_{\Gamma_1} f(z)dz$$



This allows us to compute complicated line integral by an easy line integral.

How to shows this?

Since f is holomorphic on int σ and intσ_2, by Cauchy's theorem:

$$\int_{\sigma_1} f(z)dz = 0 = \int_{\sigma_0} f(z)dz$$

So we have that:

$$\begin{aligned} 0 &= \int_{\sigma_0} f(z)dz + \int_{\sigma_1} f(z)dz \\ &= \int_{\gamma_0} f(z)dz - \int_{\gamma_1} f(z)dz + 0 \end{aligned}$$

Cauchy integral formula

Theorem 8. Suppose $D \subseteq \mathbb{C}$ is open and $f : D \to \mathbb{C}$ is holomorphic.

If $z_0 \in D$,

$$f(z_0) = \frac{1}{2\pi i} \int_\gamma \frac{f(z)}{z - z_0} dz$$

For every closed, simple curve γ inside D whose interior contains z_0 and is counter-clockwise oriented.

How to show this?

First:

$$\frac{f(z)}{z - z_0}$$

is holomorphic on $D \setminus \{z_0\}$. By extended Cauchy's theorem we have:

$$\int_\gamma \frac{f(z)}{z - z_0} = \int_{\gamma_1} \frac{f(z)}{z - z_0} dz$$

where $\gamma_1 : [0, 2\pi] \to \mathbb{C}$, $\gamma_1(t) = z_0 + re^{it}$, for a small r. Hence:

$$\begin{aligned} \int_\gamma \frac{f(z)}{z - z_0} dz &= \int_{\gamma_1} \frac{f(z)}{z - z_0} dz \\ &= \int_0^{2\pi} \frac{f\big(z_0 + re^{it}\big)}{re^{it}} ire^{it} dt \\ &= i \int_0^{2\pi} f\big(z_0 + re^{it}\big) dt \to i2\pi f(z_0) \text{ as } r \to 0 \end{aligned}$$

using that f is continuous at z_0.

2.3.2 Trick to compute line integrals easily

- Apply Cauchy theorem $\rightarrow$ find simply connex set u 'enclosing' loop γ on which f is holomorphic
- Apply extension of the Cauchy theorem $\rightarrow$ replace $\int_{\gamma_0} f dz$ over a 'very complicated' Γ_0 by $\int_{\gamma_1} f dz$ over a 'simpler' Γ_1
- If Γ_0, Γ_1 have some endpoint and orientation and f is holomorphic on the region they enclose. Then:

$$\int_{\Gamma_0} f dz = \int_{\Gamma_1} f dz$$

Extension of Cauchy's integral formula

Under the same hypothesis as for the Cauchy's integral formula:

Theorem 9.

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_\gamma \frac{f(z)}{(z - z_0)^{(n+1)}} dz$$

Remark And we can see that if $n = 0$ then this fall under the usual formula.

This gives us another way to compute a maybe 'complicated' using something that 'may' be easier.

This essentially follows by differentiating the Cauchy's integral formula

Proof. We first start from the Cauchy integral Formula:

$$f(z_0) = \frac{1}{2\pi i} \int_\gamma \frac{f(z)}{z - z_0} dz$$

We can take the derivative on both side

$$f'(z) = \frac{1}{2\pi i} \frac{d}{dz_0} \int_\gamma \frac{f(z)}{z - z_0} dz$$

Using Leibniz rule, we move the derivative inside the integral:

$$f'(z_0) = \frac{1}{2\pi i} \int_\gamma \frac{\partial}{\partial z_0} \left(\frac{f(z)}{z - z_0} \right) dz$$

Since $f(z)$ does not depend on z_0, only the denominator is differentiated, which means that we need to compute:

$$\frac{df}{dz_0} (z - z_0)^{-1} = (z - z_0)^{-2}$$

Now if we substitutes this into our equation

$$f'(z_0) = \frac{1}{2\pi i} \int_\gamma \frac{f(z)}{(z - z_0)^2}$$

Now to prove the general result (not only the base case) we will need to do the induction steps. Assume that $f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_\gamma \frac{f(z)}{(z-z_0)^{n+1}} dz$. If we retake its derivative as before we have

$$\begin{aligned} f^{(n+1)}(z_0) &= \frac{n!}{2\pi i} \frac{d}{dz_0} \int_\gamma \frac{f(z)}{(z - z_0)^{n+1}} dz \\ &= \frac{n!}{2\pi i} \int_\gamma \frac{\partial}{\partial z_0} \frac{f(z)}{(z - z_0)^{n+1}} \end{aligned}$$

Let us now compute:

$$\frac{\partial}{\partial z_0}\frac{1}{(z-z_0)^{n+1}} = --\frac{n+1}{(z-z_0)^{n+2}}$$

which gives us:

$$f^{(n+1)}(z_0) = \frac{(n+1)!}{2\pi i}\int_\gamma \frac{f(z)}{(z-z_0)^{(n+1)+1}}dz$$

□

Remark 11. This formula is used to prove a statement that we have already seen: holomorphic functions are infinitely many times differentiable.

Example

Let $f(z) = \frac{\cos(z)}{z}$ and let γ be a simple closed curve, oriented counter-clockwise. The question that we are trying to answer is: what are the value of $\int_\gamma f dz$?

Before starting anything we can see our function kind look like what is being integrated above. Knowing that we can first check some information about our function:

- If $0 \in \gamma$ this integral is ill-defined
- If $0 \notin \gamma$ and 0 is not in the interior of γ then we have that our function is holomorphic even inside the curve. Therefore by Cauchy integral theorem:

$$\int_\Gamma f dz = 0$$

- If $0 \notin \gamma$ yet 0 is in the interior of γ then we note

$$\int_\gamma f(z)dz = \int_\gamma \frac{g(z)}{z}dz$$

By the Cauchy integral formula for $g(z) = \cos(z)$ and $z_0 = 0$ this gives us

$$= 2\pi i g(0) = 2\pi i$$

Example II

Let γ be the circle $\{z \in \mathbb{C} : \|z-2\| = 1\}$ oriented counter-clockwise. Compute $\int_\gamma \frac{e^z}{(z-2)^3}dz$.

This is a nice function under a power... First we can notice that e^z is holomorphic on $\mathbb{C}$ therefore, let us developed using the formula

$$= \int_\gamma \frac{f(z)}{(z-2)^{2+1}} = \frac{2\pi i}{2!}f(2) = \pi i e^2$$

Chapter 3

Laurent Series

Before jumping to Laurent series, first let's recall what Taylor series are and why are we interested in them.

3.1 Taylor series

The goal of Taylor series is to approximate any function using polynomial. The reason for this is that a polynomial is very nice to manipulate (derivative/primitive are very easy to compute) the function is continuous etc...

Definition 19. Taylor Series Let $f : \mathbb{R} \to \mathbb{R}$ sufficiently regular function. For any $x_0 \in \mathbb{R}$ the **Taylor polynomial** of degree N is

$$T_{x_0}^N f(x) = f(x_0) + f'(x_0)(x - x_0) + \cdots + \frac{f^{(n)(x_0)}}{N!}(x - x_0)^N$$

Theorem 10. If $f \in C^{N+1}(\mathbb{R})$ then for all $x_1, x_0 \in \mathbb{R}$ there exists $\theta \in (0, 1)$ such that

$$f(x) = T_{x_0}^N f(x) + \frac{f^{N+1}(\theta x_0 + (1 - \theta)x)}{(N + 1)!}(x - x_0)^{N+1}$$

In practice, θ is not known, but one can usually estimate the $N + 1$ derivative. Also, Taylor's theorem informally means that we can approximate function by polynomials.

Definition 20. If f has derivatives of all orders, then we may consider the **Taylor series**

$$\sum_{n=0}^{\infty} \frac{f^{(n)(x_0)}}{n!}(x - x_0)^n$$

An important question which comes up using series is: Does this series actually converge? If we take for instance $f(x) = e^x, x_0 = 0$. The Taylor series around x_0 is given as

$$\sum_{n=0}^{\infty} \frac{1}{n!} x^n$$

In this case is converges everywhere on $\mathbb{R}$ by definition of the exponential function.

A bad example of convergence this time would be $f(x) = \frac{1}{1-x}, x = 0$. This Taylor series

around x_0 is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}}{n!}(x-0)^n = \sum_{n=0}^{\infty} x^n$$

since $f^{(n)}(x) = (1-x)^{-n-1}$.

This series on the right hand side converges for $\|x\| < 1$.

Remark 12. Functions f such that the Taylor series converges to f in a neighbourhood of x_0 are called **analytic**. Not all smooth function are analytic, e.g.

$$f(x) = \begin{cases} e^{-\frac{1}{x}} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

is smooth but not analytic.

3.2 Taylor series/ Regular Laurent Series for complex functions

Taylor series or more specifically regular Laurent series (which we will see later in this course).

Theorem 11. Let $D \subseteq \mathbb{C}$ be open and $z_0 \in D$. Let $R > 0$ be such that the disk

$$B_R(z_0) := \{z \in \mathbb{C} : \|z - z_0\| < R\}$$

Is contained in D.

If $f : D \to \mathbb{C}$ is holomorphic then

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)(z_0)}}{n!}(z-z_0)^n$$

For all $z \in B_r(z_0)$

What this means is that if I gave you a point and a ball around that point. Then, at each point around this ball the function at this point is equal to his Taylor series.

The largest possible R is called the **radius of convergence**

Remarks

1. In particular, all holomorphic functions are analytic
2. In practice: R is the distance of z_0 from the nearest '*point*' where f is not holomorphic/defined
3. We can connect the Taylor series to Cauchy's integral formula. This would say that f can be written as a power series

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}}{n!}(z-z_0)^n = \sum_{n=0}^{\infty} c_n (z-z_0)^n$$

Where

$$c_n = \frac{f^{(n)(z_0)}}{n!} = \frac{1}{2\pi i}\int_\gamma \frac{f(\zeta)}{-z_0} d\zeta$$

For every simple, closed curve γ around z_0 with int $\gamma \subseteq D$ and counter-clockwise

Example

Let $f(z) = e^z$ is holomorphic on $\mathbb{C}$. Therefore for every $z_0 \in \mathbb{C}$ the theorem applies for any $R > 0$ (so f has infinite radius of convergence) and

$$e^z = \sum_{n=0}^{\infty} \frac{e^{z_0}}{n!}(z - z_0)^n$$

Example II

Let $f(z) = \frac{1}{1-z}$ holomorphic on $D = \mathbb{C} \setminus \{1\}$. Let $z_0 = 0$. In this case, the largest possible R is 1. For $z \in B_1(0)$ this gives us

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} z^n = \sum_{n=0}^{\infty} z^n$$

Since $f^n(0) = n!$

Example III

Let $f(z) = \frac{1}{1+z^2}$ holomorphic on $D = \mathbb{C} \setminus \{-i, i\}$. Let $z_0 = 0$ which again give us as the largest R, 1. Using $B_1(0)$, we could simply use

$$\frac{1}{1-w} = \sum_{n=0}^{\infty} w^n$$

And insert $w = -z^2$ to get:

$$\frac{1}{1+z^2} = \sum_{n=0}^{\infty} (-1)^n z^{2n}$$

Caution

The Taylor series representation around a given point is unique.

In particular, in order to compute the Taylor series representation at a given point, we don't have to do the 'straightforward' computation, we can use trick by using other Taylor series that we already know. This is something that we have already done in real analysis (for instance when doing limit of function of the type $e^{\sin(x)}$)

3.2.1 Laurent Series

The intuition behind this is, can we change in a way our series so that we can represent function even when the Taylor series is not defined. To do so we will allows our series to have **negative** exponent.

Theorem 12. Let $D \subseteq \mathbb{C}$ be open and $z_0 \in D$. Assume $f : D \setminus \{z_0\} \to \mathbb{C}$ is holomorphic. Let $R > 0$ such that $B_R(z_0) \subseteq D$ Then for all $z \in B_R(z_0) \setminus \{z_0\}$

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n$$

Where for every simple closed, counter-clockwise oriented curve γ such that int $\gamma \subseteq D$ and z_0 is enclosed by γ,

$$c_n = \frac{1}{2\pi i} \int_\gamma \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta$$

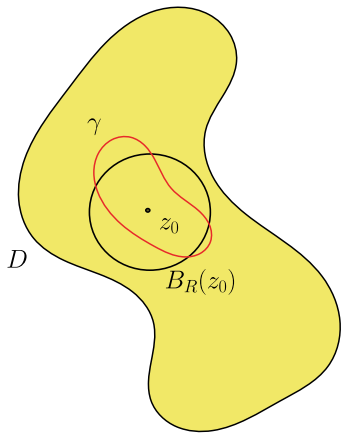


Figure 3.1: representation

Intuition The Laurent Series is like a Taylor series but now we include terms with polynomial singularities.

And as for the Taylor series, the Laurent series representation is **unique**.

If we find the coefficients c_n by any means (not necessarily by a path integral), then this has to be the Laurent series.

Definition 21.
Any Laurent Series splits into a **singular** and **regular** part

$$\sum_{n=-\infty}^{\infty} c_n (z - z_0)^n = \underbrace{\cdots + \frac{c-2}{(z-z_0)^2} + \frac{c-1}{z-z_0}}_{\text{singular part}} + \underbrace{c_0 + c_1(z - z_0) + \cdots}_{\text{regular part}}$$

In other words, the Taylor series is a Laurent series without singular part.

Example

If $f : D \to \mathbb{C}$ holomorphic and $z_0 \in D$ then the Taylor series implies that

$$f(z) = \sum_{n=0}^{\infty} x_n (z - z_0)^n = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n$$

The singular part of the Laurent series has to vanish since for all $n \leq -1$ we have

$$\frac{f(\zeta)}{(\zeta - z_0)^{n!1}} \text{ is holomorphic near } z_0$$

By Cauchy's integral formula $\int_\gamma \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d = 0$

Example II

For the following function $f : \mathbb{C} \setminus \{0\} \to \mathbb{C}$ let us find the Laurent series around $z_0 = 0$

1. $f(z) = \frac{1}{z}$ This is already a Laurent series

$$f(z) = \sum_{n=-\infty}^{\infty} c_n z^n \qquad \begin{cases} 1 & n = -1 \\ 0 & \text{otherwise} \end{cases}$$

2. $f(z) = \frac{5z+1}{z^2} = \frac{1}{z^2} + \frac{5}{z}$ which we can again translate into a Laurent series:

$$= \sum_{n=-\infty}^{\infty} c_n z^n$$

Where

$$c_n = \begin{cases} 1 & n = -2 \\ 5 = n = -1 \\ 0 & \text{otherwise} \end{cases}$$

Remark 13. The Laurent series is unique. Hence, if we manage to compute a power series expansion for f as above by any means, (e.g. from another Laurent series), we have determined the Laurent series.

Example

If we know that:

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n \qquad \text{for } |z| < 1$$

Then if we want to determine the Laurent series of $\frac{1}{1+z^2}$ we could have:

$$\frac{1}{1+z^2} = \frac{1}{1-(-z^2)} \underbrace{=}_{|1|<1} \sum_{n=0}^{\infty} \left(-z^2\right)^n$$
$$= \sum_{n=0}^{\infty} c_k z^k$$

Where

$$c_k = \begin{cases} 0 & k \text{ odd} \\ (-1)^{\frac{k}{2}} & k \text{ even} \end{cases}$$

Example II

Let $f : \mathbb{C} \setminus \{0, -1\} \to \mathbb{C}$, $f(z) := \frac{1}{z(z+1)}$. What we are trying to do is to develop the Laurent series of f at a given point $z_0 \in \mathbb{C}$. In practice, the way to go is to look at the function, to see where it is behaving nicely or not. But here as we don't know (for an arbitrary point z_0) we need three cases:

1. Assume $z_0 = 0$, we can observe that $f(z) = \frac{1}{z} - \frac{1}{z+1}$. But here we just summoned a known Taylor series $\frac{1}{z-(-1)}$ Which gives us

$$\frac{1}{1-(-z)} = \sum_{n=0}^{\infty} (-1)^n z^n \qquad \text{where } |z| < 1$$

So $f(z) = \frac{1}{z} - \sum_{n=0}^{\infty}(-1)^n z^n$ But here the $\frac{1}{z}$ is just a term of the regular part of the Laurent series, therefore this gives us:

$$= \sum_{n=-1}^{\infty} (-1)^{n+1} z^n$$

The radius of convergence for this series is give by $R = 1$, it is the distance to z_0 to the next distinct singular point of f(which is given by $z = -1$).

2. Assume $z_0 = -1$. Using the same trick as in (1): $f(z) = -\frac{1}{z-(-1)} + \frac{1}{z}$. Now if we expand $\frac{1}{z}$ as Taylor series around -1:

$$\frac{1}{z} = \sum_{n=0}^{\infty}(-1)(z-(-1))^n \quad \text{where } |z-(-1)| < 1$$

so

$$\begin{aligned} f(z) &= -\frac{1}{z-(-1)} + \sum_{n=0}^{\infty}(-1)(z-(-1))^n \\ &= \sum_{n=-1}^{\infty} (-1)(z-(-1))^n \end{aligned}$$

With a Radius of converge of 1

3. Assume now that $z_0 \neq 0, -1$. In this case, f is holomorphic on an open set containing z_0 so around z_0 f can be written as a regular Taylor series. This series converges in any disk around z_0 which does not contain 0 or -1. The radius is given by $R = \min\{|z_0|, |z_0+1|\}$ Does are the distance between our point z_0 and the point 0 and -1.

The principle is to use as much knowledge as we know about Taylor/Laurent series and common functions in order to compute new series.

Regular point

Definition 22. Regular point

We say f has a **regular point** at z_0 if the singular part of the Laurent series at z_0 vanishes

2. We say f has a **pole of order m** at z_0 if the coefficients of the Laurent series of f at z_0 satisfy $c_n = 0 \forall n < -m$, i.e.

$$f(z) = \frac{c_m}{(z-z_0)^m} + \frac{c_{m+1}}{(z-z_0)^{m-1}} + \cdots, \quad c_m \neq 0$$

3. We say f has an essential singularity at z_0 if its Laurent series has infinitely many coefficients $c_n \neq 0$ with $n \leq 1$

The intuition behind the second point if that '*f blows up like $\frac{1}{(z-z_0)^m}$ at z_0*'

Example Let $f(z) = \sum_{n=0}^{\infty} \frac{z^n}{n!} = e^z$, here 0 is regular.

Remark If f is holomorphic at z_0, then z_0 is a regular point. The converse is also true: if z_0 is a regular point of f, then

$$\lim_{z \to z_0} f(z)$$

Exists and if we extend f to z_0 this way, f is holomorphic at z_0. In short

$$z_0 \text{ is regular point of } f \iff \lim_{z \to z_0} f(z) \text{ exists}$$

Example II

- Let $f(z) = \frac{1}{z}$ has a pole of order 1 at $z_0 = 0$
- $f(z) = \frac{1}{(z+2)^2} + \frac{1}{z}$ has a pole of order 2 at $z_0 = -2$ and a pole of order 1 at $z_0 = 0$

The second statement comes from the fact that the function can be directly written as a Laurent series with the squared represent the second part of the Laurent series which implies an order of 2 this part will only contribute to the Taylor series (because this function is holomorphic). $\frac{1}{z}$ will definitely belongs to the singular part.

If f has a pole of order m at z_0, then we can write $f(z)$ to be:

$$f(z) = \frac{c_{-m}}{(z-z_0)^m} + \frac{c_{-m+1}}{(z-z_0)^{m-1}} + \cdots$$

Which means that if we take for instance $g(z) = (z-z_0)^m f(z)$ has a regular point at z_0

In fact f has a pole of order m at z_0 if and only if $\lim_{z\to z_0}(z-z_0)^m f(z)$ exists and is non-zero.

Which means that we can multiply our function by a power of z and the function will still exists (which is kind of obvious) so for instance $x^{2000} \cdot \frac{1}{z}$ when going to 0 the limit will exists and will be 0 (because the order is 1 and not 2000). In particular if f has a pole of order m at z_0, we can write

$$f(z) = \frac{g(z)}{(z-z_0)^m}$$

where g is holomorphic at z_0 and $g(z_0) \neq 0$ (the opposite is also true)

Remark 14. The fact that we can rewrite f as a product $\frac{g(z)}{z-z_0}$ for a given singularity is pretty big. Using Cauchy's Integral Formula, this implies that if $z_0 \in \gamma$ and that $g(z)$ is analytic inside and on γ, then:

$$\oint_\gamma \frac{g(z)}{z-z_0} dz = 2\pi i \cdot g(z_0)$$

This means that we can solve compute integral of another function to get the integral of our function f.

Example III

Let us for instance consider $f(z) = e^{\frac{1}{z}}$ at $z_0 = 0$. We can use the Taylor series of the exponential in order to compute the Laurent series:

$$f(z) = \sum_{n=0}^{\infty} \frac{\left(\frac{1}{z}\right)^n}{n!} = \sum_{n=-\infty}^{0} \frac{1}{(-n)!} z^n$$

So 0 is an essential singularity of f.

Example III ii

Let $f(z) = ze^{\frac{1}{z}}$ at $z_0 = 0$. Using the same trick as above

$$f(z) = z \sum_{n=-\infty}^{0} \frac{1}{(-n)!} z^n = \underbrace{z+1}_{\text{regular part}} + \underbrace{\frac{1}{2!z} + \frac{1}{3!z^2} + \cdots}_{\text{singular part}}$$

So 0 is an essential singularity

Now let's discuss quotients of Taylor series in this context. Suppose $p, q : D \setminus \{z_0\} \to \mathbb{C}$ holomorphic with z_0 being regular for both function. Regular means that we can write the Laurent series as:

$$p(z) = a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots$$

$$q(z) = b_0 + b_1(z - z_0) + b_2(z - z_0)^2 + \cdots$$

The goal is to study singularities of the quotients $\frac{p}{q}$ near z_0. In particular we are asking ourself what is the order of the pole (if there is any)? Let's discuss a few instructive examples. What we will harm here is our q, we will need to be careful when $q(z) = 0$, we will need to do some assumptions:

1. if $b_0 \neq 0$ so $q(z_0) \neq 0$. In particular $\frac{1}{q}$ is holomorphic near z_0 and we have that

$$\lim_{z \to z_0} \frac{p(z)}{q(z)} = \frac{p(z_0)}{q(z_0)} = \frac{a_0}{b_0}$$

   exists which implies that z_0 is a regular point for $\frac{p}{q}$
2. If $a_0 = b_0 = 0$ and this case both function vanish at 0... To make our life easier let us assume that $b_1 \neq 0$ then

$$\begin{aligned} \lim_{z \to z_0} \frac{p(z)}{q(z)} &= \lim_{z \to z_0} \frac{a_1(z - z_0) + a_2(z - z_0)^2 + \cdots}{b_1(z - z_0) + b_2(z - z_0)^2 + \cdots} \\ &= \lim_{z \to z_0} \frac{a_1 + a_2(z - z_0) + \cdots}{b_1 + b_2(z - z_0) + \cdots} = \frac{a_1}{b_1} \end{aligned}$$

   because the limit exists, z_0 is a regular point for f.

   Similarly if $a_0 = b_0 = 0$ and $a_1 = b_1 = 0$ yet $b_2 \neq 0$.

Example

Let our function be $f(z) = \frac{z}{\sin z}$ where $z_0 = 0$. Using power series expansion of the sin function, we can convince ourself that the limit of z approaching 0 does exists (recall analysis I):

$$f(z) = \frac{z}{z - \frac{z^3}{3!} + \frac{z^5}{5!} + \cdots} = \frac{1}{1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \cdots}$$

By setting $z = 0$ we can see that the limit does exists:

$$\lim_{z \to 0} f(z) = 1$$

Therefore, 0 is a regular point

Summary In Summary, in order to see if the limit exists, either expand in power series and take out as many common powers of $z - z_0$ from the nominator and denominator as possible and take the limit, or use l'Hospital's rule.

Remark 15. This looks a lot like how we used to compute limit in analysis I and yes it is, We use the same principle here than there.

Example

Suppose that

$$p(z) = a_k(z-z_0)^k + a_{k+1}(z-z_0)^{k+1} + \cdots, \quad a_k \neq 0$$
$$q(z) = b_l(z-z_0)^l + b_{l+1}(z-z_0)^{l+1} + \cdots, \quad b_l \neq 0$$

Consider now the function $f(z) = \frac{p(z)}{q(z)}$ near z_0.

- If $k \geq l$ then z_0 is a regular point for f. To do so we will show that the limit of our function when approaching the point converges

$$\lim_{z\to z_0} f(z) = \lim_{z\to z_0} \frac{a_k(z-z_0)^k + a_{k+1}(z-z_0)^{k+1} + \cdots}{b_l(z-z_0)^l + b_{l+1}(z-z_0)^{l+1} + \cdots}$$

  Because $k \geq l$, we can factor out $(z-z_0)^l$ which gives us

$$= \lim_{z\to z_0} \frac{a_k(z-z_0)^{k-l} + a_{k+1}(z-z_0)^{k+1-l} + \cdots}{b_l + b_{l+1}(z-z_0) + \cdots}$$
$$= \begin{cases} 0 & \text{if } k > l \\ \frac{a_k}{b_l} & \text{if } k = l \end{cases}$$

- If $k < l$ then z_0 is a pole of order $l-k$. Using the same trick as what we have done above, factoring out until we see holomorphic function:

$$\begin{aligned} f(z) &= \frac{a_k(z-z_0)^k + a_{k+1}(z-z_0)^{k+1} + \cdots}{b_l(z-z_0)^l + b_{l+1}(z-z_0)^{l+1} + \cdots} \\ &= \frac{a_k + a_{k+1(z-z_0)+\cdots}}{b_l(z-z_0)^{l-k} + b_{l+1}(z-z_0)^{l+1-k} + \cdots} \\ &= \underbrace{\frac{1}{(z-z_0)^{l-k}}}_{\text{blows up at } z_0} \left[\frac{a_k + a_{k+1}(z-z_0) + \cdots}{b_l + b_{l+1}(z-z_0) + \cdots}\right] \end{aligned}$$

  This right part is holomorphic at z_0 Laurent series with only regular part. Therefore, we could write our function:

$$f(z) = \frac{c_0}{(z-z_0)^{l-k}} + \frac{c_1}{(z-z_0)^{l-k+1}} + \cdots$$

Example Let $f(z) = \frac{\sin z}{2z^3+z^4}$ has a pole of order 2 at $z_0 = 0$.

Since $\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots$ we have that

$$\begin{aligned} f(z) &= \frac{z - \frac{z^3}{z!} + \frac{z^5}{5!} - \cdots}{2z^3 + z^4} = \frac{1 - \frac{z^2}{3!} + \frac{z^4}{5!} + \cdots}{2z^2 + z^3} \\ &= \frac{1}{z^2}\left[\frac{1 - \frac{z^2}{3!} + \frac{z^4}{5!}}{2+z}\right] \end{aligned}$$

Example

Let $f(z) = z\sin\left(\frac{1}{z}\right)$, the claim is that this function has an essential singularity at 0.

Indeed, we can realise this by using the Taylor expansion for this sin function which will gives us a Laurent series:

$$\sin\frac{1}{z} = \sum_{n=0}^{\infty} \frac{(-1)^k}{(2n+1)!}\left(\frac{1}{z}\right)^{n+1}$$

And hence, by multiplying this sum by z:

$$z \sin \frac{1}{z} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \frac{1}{z^{2n}}$$

And this indeed converges when $z \neq 0$.

3.3 Residues

A residue is a very simple object: it is the first coefficient of the Laurent series

Definition 23. Residue The coefficient c_{-1} of the term $(z - z_0)^{-1}$ in the Laurent series for f at z_0 is called the **residue** of f at z_0. Denoted by:

$$\mathrm{Res}_{z_0}(f) := c_{-1}$$

Remark 16. We have already seen that the coefficients of the Laurent series can be described directly using line integral.

$$\mathrm{Res}_{z_0}(f) = \frac{1}{2\pi i} \int_\gamma f(z) dz$$

Where γ is a closed, simple, counterclockwise oriented curve that contains z_0 in its interior and f is holomorphic on the int $\gamma \setminus \{z_0\}$

Now let us compute some residue!!!!!!!!!!!!!!!!!!

Example

If f is holomorphic on an open set containing z_0,

$$\mathrm{Res}_{z_0}(f) = 0$$

By Cauchy's theorem

The concept of residue becomes interesting when we have singularity inside our function, especially at 0. For instance if we take:

$$\mathrm{Res}_{z_0}\left(\frac{1}{z}\right)$$

Then we can see that we already have the Laurent series of $\frac{1}{z}$ at the point 0.

$$\mathrm{Res}_{z_0}\left(\frac{1}{z}\right) = 1$$

Suppose more generally f has a pole of order m at z_0, i.e. its Laurent series reads

$$f(z) = c_{-m}(z - z_0)^{-m} + \cdots + c_{-1}(z - z_0)^{-1} + c_0 + \cdots$$

Where $c_{-m} \neq 0$, consider the holomorphic function

$$\begin{aligned} g(z) &= (z - z_0)^m f(z) \\ &= c_{-m} + c_{-m+1}(z - z_0) + \cdots + c_{-1}(z - z_0)^{m-1} + \cdots \end{aligned}$$

Then we can extract the residue by the formula

$$\begin{aligned} c_{-1} &= \frac{1}{(m-1)!}\frac{d^{m-1}}{dz^{m-1}}\Big|_{z_0}(z) \\ &= \lim_{z\to z_0}\frac{1}{(m-1)!}\frac{d^{m-1}}{dz^{m-1}}((z-z_0)^m f(z)) \end{aligned}$$

This formula is useful if we already know that f has a pole of order m at z_0. In particular if $m = 1$ then this becomes

$$\mathrm{Res}_{z_0}(f) = \lim_{z\to z_0}(z-z_0)f(z)$$

Example

Let us consider $f(z) = \frac{1}{\sin z}$ at $z_0 = 0$ which has a pole of order 1:

$$f(z) = \left[z - \frac{z^3}{3!} + \cdots\right]^{-1} = \frac{1}{z}\underbrace{\left[1 - \frac{z^2}{3!} + \cdots\right]^{-1}}_{\text{regular}}$$

Hence from the above example,

$$\begin{aligned} \mathrm{Res}_0(f) &= \lim_{z\to 0} zf(z) \\ &= \lim_{z\to 0}\frac{z}{\sin z} \\ &= \lim_{z\to 0} z\left[z - \frac{z^3}{3!} + \cdots\right]^{-1} \\ &= \lim_{z\to 0}\left[1 - \frac{z^2}{3!} + \cdots\right]^{-1} = 1 \end{aligned}$$

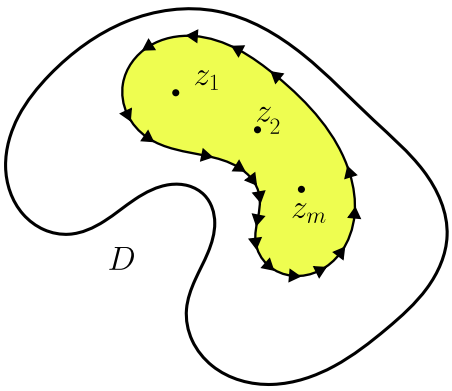
3.3.1 Residue Theorem

Theorem 13. Suppose $D \subseteq \mathbb{C}$ is an open and simply connected set and let γ be a simple closed curve in D which is counterclockwise oriented. Let $z_1, \ldots, z_m \in \ \mathrm{int}\gamma$.

Assume $f : D \setminus \{z_1, \ldots, z_m\} \to \mathbb{C}$ is holomorphic

then

$$\int_\gamma f(z)dz = 2\pi i[\mathrm{Res}_{z_1}(f) + \cdots + \mathrm{Res}_{z_m}(f)]$$

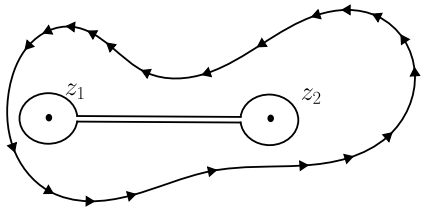


Hence, in order to compute the line integral of $\int_\gamma f dz$ it suffices to compute the residues of the singular points in the interior of γ.

How does one show this?

For simplicity, we will consider for 2 points only (the same argument foes for n points)

What we are doing geometrically is creating two curves ρ_1, ρ_2 that turns around z_1 and z_2 and join them together with a very small line, geometrically this gives us something like this:



By the extension of Cauchy's theorem

$$\int_\gamma f dz = \int_\rho f dz$$

Then we are trying to see what happens with the integral of ρ when the radius becomes very thin. First, for the '*bridge*', as it becomes very narrow, it will cancel out. This leaves us with the integral over ρ_1, ρ_2. Therefore

$$\int_\gamma f dz = \int_{\rho_1} f dz + \int_{\rho_2} f dz = 2\pi i \mathrm{Res}_{z_1}(f) + 2\pi i \mathrm{Res}_{z_2}(f)$$

Remark Recall that:

$$\int_{\rho_1} f dz = 2\pi i \mathrm{Res}_{z_1}(f)$$

Comes down from the definition of the Residue seen in the Remark 14.

Example

Consider $f(z) = \frac{1}{z}$. Let γ be a simple closed counterclockwise curve in $\mathbb{C}$. Compute $\int_\gamma f dz$

1. If $0 \in \text{int } \gamma$ then by the residue theorem

$$\int_\gamma f dz = 2\pi i Res_0(f) = 2\pi i$$

2. If $0 \notin \text{int } \gamma \cup \gamma$ then

$$\int_\gamma f dz = 0$$

3. if $0 \in \gamma$ then

$$\int_\gamma f dz \text{ is not well-defined}$$

Example

Let $f : \mathbb{C} \setminus \{0, -1\} \to \mathbb{C}$, $f(z) = \frac{1}{z(z+1)}$. We have already seen f has two poles of order 1 at $z_0 = 0$ and $z_1 = -1$. Hence

Let γ be a closed counterclockwise oriented curve. The queis always the same: what is $\int_\gamma f dz$?

1. $0, -1 \notin$ int $\gamma \cup \gamma$ then $\int_\gamma f dz = 0$
2. $0 \in$ intγ but $-1 \notin$ int $\gamma \cup \gamma$ then
$$\int_\gamma f dz = 2\pi i \text{Res}_0(f) = 2\pi i$$
3. $0 \notin$int $\gamma \cup \gamma$ but $-1 \in$ int γ then
$$\int_\gamma f dz = 2\pi i \text{Res}_{-1}(f) = -2\pi i$$
4. If $0, -1 \in$ int γ then by the residue theorem
$$\int_\gamma f dz = 2\pi i[\text{Res}_0(f) + \text{Res}_{-1}(f)] = 0$$
5. $0 \in \gamma$ or $-1 \in \gamma$ then the integral is not well defined

What is important is that when we are asked to compute simple line integral, it is important the divide cases based the singularities inside of γ

Example

Let $f : \mathbb{C} \setminus \{0, 1\} \to \mathbb{C}$, $f(z) = \frac{1}{z^2} + \frac{2}{z} + \frac{3}{z-1}$. This function has a pole of order 2 at 0 and a pole of order 1 at 1

Example

Let $\exp\left(\frac{1}{z^2}\right) = 1 = \frac{1}{z^2} + 1\frac{1}{2!}\frac{1}{z^2} + \cdots$ hence
$$\text{Res}_0\left(\exp\left(\frac{1}{z^2}\right)\right) = 0$$

3.3.2 Applications

The residue theorem can be used to compute integrals over simple closed curve, sometimes even to compute **real** integral by replacing them with complex integrals.

Example

Let us for instance compute
$$\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$$

If you have forgotten that this is the primitive of arctan then you can still try to rephrase this into a complex integral. Let us set $f(z) = \frac{1}{1+z^2}$.

Given $R > 0$, consider the segment $L_R : [-R, R] \to \mathbb{C}$ given by $L_R(t) = t$ and the half circle $c_R : [0, \pi] \to \mathbb{C}$ given by $C_R(t) = Re^{it}$

Consider the closed curve $\gamma = LR \cup C_R$. Then the function $f(z) = \frac{1}{1+z^2}$ has

poles of order 1 at i. From the residue theorem then

$$\int_\gamma f dz = 2\pi i \mathrm{Res}_i(f)$$

But we can compute the residue explicitly which is given by

$$\mathrm{Res}_i(f) = \lim_{z\to i}(z-i)f(z) = \lim_{z\to i}\frac{1}{z+i} = \frac{1}{2i}$$

So:

$$\begin{aligned}\int_\gamma ddz = \pi &= \int_{L_R} f dz + \int_{C_R} f dz \\ &= \int_{-R}^{R}\frac{1}{1+t^2}dt \underbrace{\to}_{R\to\infty} \int_{-\infty}^{\infty}\frac{1}{1+t^2}dt\end{aligned}$$

On the other hand,

$$\int_{C_R} f dz = \int_0^\pi \frac{1}{1+R^2e^{2it}} Rie^{it}dt \approx \frac{\pi}{R}\underbrace{\to}_{R\to\infty}$$

This method works for any integrand that decays faster than $\frac{1}{R}$ as $R\to\infty$

Example

Let $R(x) = \frac{P(x)}{Q(x)}$ where P, Q are real polynomials and deg $Q \geq$ deg $P + 2$. Assume R is defined on $\mathbb{R}$ which means that $Q(x) \neq 0 \quad \forall x \in \mathbb{R}$

Observations

1. The latter condition on Q forces deg Q to be even
2. Q has real coefficients if $Q(z) = 0$ for $z \in \mathbb{C}$ then by taking the complex conjugate of this equations, we find $Q(\overline{z}) = 0$ So roots of Q in $\mathbb{C}$ come in pairs of complex conjugates (none of which are real)

Given $a \geq 0$ we want to compute

$$\int_{-\infty}^{\infty} R(x)\cos(ax)dx, \quad \int_{-\infty}^{\infty} R(x)\sin(ax)dx$$

By Euler's formula $e^{iax} = \cos(ax) + i\sin(ax)$, we can compute both by computing

$$\int_{-\infty}^{\infty} R(x)e^{iax}$$

Remark And then we can read off the real and imaginary part of the result. Such integrals appear often in applications (e.g. Fourier transform, resonances in harmonic systems with friction, etc.).

Consider $f(z) = R(z)e^{iaz}$, let $z_1, \ldots, z_k$ be zeros of Q, then:

$$f : \mathbb{C}\setminus\{z_1,\ldots,z_k\} \to \mathbb{C}$$

is holomorphic.

The goal now will be to use the same trick as before, making our path be a half circle in the upper complex plane and a line on the Real axis, but this time we need a r large enough so that $z_1, \ldots, z_k$ are contained in the disk $B_r(0)$.

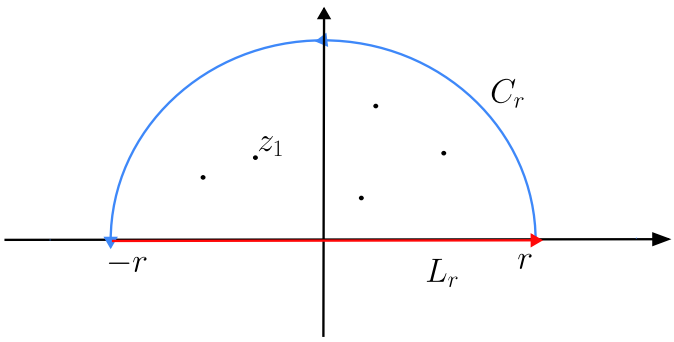


Figure 3.2: Path $C_r + L_r$

As we can see above, we can construct our two paths:

$$L_r = \{z \in \mathbb{C} : \text{Im } z = 0, -r \leq \text{Re } z \leq rz\}$$

parametrized by $L_r(t) = t$, $t \in [-r, r]$

$$C_r = \left\{re^{i\theta} : \theta \in [0, \pi]\right\}$$

Now we can defined the whole path:

$$\Gamma_r := L_R \bigcup C_r$$

and the integral over this path:

$$\int_{\Gamma_r} f dz = \underbrace{\int_{L_r} f dz}_{=\int_r^r R(t)e^{iat}dt} + \int_{C_r} f dz$$

As $r \to \infty$, one can compute $\int_{\Gamma_r} f dz$ by using the residue theorem: if $z_1, \ldots, z_k$ are zeros of Q in the upper half of the plane (which are half of the total number of zeroes of Q) then:

$$\int_{\Gamma_r} f dz = 2\pi i[\text{Res}_{z_1}(f) + \cdots + \text{Res}_{z_m}(f)]$$

As $r \to \infty$ this goes to $\int_{-\infty}^{\infty} R(x)e^{iax}dx$.

It remains to show that $\int_{C_r} f dz \to 0$ as $r \to \infty$.

- For Im $z \geq 0$ and $a \geq 0$, $\left|e^{iaz}\right| \leq 1$. Indeed, if we split z as $z = x + iy$ we can see:

$$\left|e^{ia(x+iy)}\right| = \left|e^{aix}e^{-ay}\right| = e^{-ay} \leq 1$$

Remark $\left|e^{ait}\right|$ is not bounded on all of $\mathbb{C}$, here it is because of the fact $i^2 < 0 \implies e^{\text{something } \leq 0} \leq 1$ But if Im $z < 0$ then with have a positive exponent to our number $\implies$ explodes

- $|R(z)|$ decays to 0 as $|z| \to \infty$

$$\begin{aligned}
\left|\int_{C_r} f dz\right| &= \left|\int_0^\pi f\left(re^{i\theta}\right) ire^{i\theta} d\theta\right| \\
&\leq \int_0^\pi \left|f\left(re^{i\theta}\right)\right| \left|rie^{i\theta}\right| d\theta \\
&= \int_0^\pi \frac{\left|P\left(re^{i\theta}\right)\right|}{\left|Q(re^{i\theta})\right|} \left|e^{iare^{i\theta}}\right| rd\theta
\end{aligned}$$

The first inequality comes from the triangle inequality. Then using the same fact as on the first point, $\left|e^{iare^{i\theta}}\right| \leq 1$ since $\text{Im}\left(re^{i\theta}\right) \geq 0$ for $\theta \in [0, \pi]$

To see where the part involving $\frac{P}{Q}$ tends to, not if

$$\left.\begin{aligned} P(z) &= b_l z^l + b_{l-1} z^{l_1} + \cdots \\ Q(z) &= a_k z^k + a_{k-1} z^{k-1} + \cdots \end{aligned}\right\} \frac{P(z)}{Q(z)} = \frac{1}{z^{k-l}} \frac{b_l + \frac{b_{l-1}}{z} + \cdots}{a_k + \frac{a_{k-1}}{z} + \cdots}$$

Hence

$$\left|\frac{P\left(re^{i\theta}\right)}{Q\left(re^{i\theta}\right)}\right| \leq \frac{\text{Const}}{r^{k-l}} \text{ for large } r$$

And thus:

$$\left|\int_{C_r} f dz\right| \leq \frac{\text{const}\pi}{r^{k-l}} r \to 0 \text{ as } r \to \infty$$

In summary we deduce:

$$\boxed{\int_{-\infty}^{\infty} R(x) e^{iax} dx = 2\pi i[\text{Res}_{z_1}(f) + \cdots + \text{Res}_{z_m}(f)]}$$

Where $z_1, \ldots, z_m$ are zeros of Q in the upper half of the plane and $f(z) = R(z)e^{iaz}$

Benefit One can compute the residues of f by finding the poles of it and their orders, without computing Laurent series!

Example

We want to compute the integral:

$$\int_{-\infty}^{\infty} \frac{x^2}{16 + x^4} \cos x dx$$

This is of the shape form the previous example with $a = 1$, $P(x) = z^2$, $Q(x) = 1 + x^4$, Thus the integral coincides with:

$$\int_{-\infty}^{\infty} \frac{x^2}{16 + x^4} \cos x dx = \text{Re}\left(\int_{-\infty}^{\infty} \frac{x^2}{16 + x^4} e^{ix} dx\right)$$

The singularities of $f(z) = \frac{z^2}{16+z^4}$ correspond to the complex root of $16 + z^4 = 0 \implies r = 2, \theta = \frac{\pi}{4} + k\frac{\pi}{2}$ where $k \in \{0, \ldots, 3\}$

Therefore we have our four singularities:

$$z_1 = \sqrt{2} + \sqrt{2}iz_2 = -\sqrt{2} + \sqrt{2}iz_3 = -\sqrt{2} - \sqrt{2}iz_4 = \sqrt{2} - \sqrt{2}i$$

Hence we define Q as

$$Q(z) = (z - z_1)(z - z_2)(z - z_3)(z - z_4)$$

So

$$R(z) = \frac{P(z)}{Q(z)} = \frac{z^2}{(z - z_1)(z - z_2)(z - z_3)(z - z_4)}$$

Where each z_i is a pole of order 1.

Only the z_i with an Im ≥ 0 are relevant for the integral which implies that we only care for the Residue of z_1 and z_2:

$$\mathrm{Res}_{z_1}(f) = \lim_{z \to z_1} (z - z_1) f(z) = \frac{e^{-\sqrt{2}} e^{\sqrt{2}i}}{4\sqrt{8}} \frac{1+i}{i}$$
$$\mathrm{Res}_{z_2}(f) = \lim_{z \to z_2} (z - z_2) f(z) = \frac{e^{-\sqrt{2}} e^{-\sqrt{2}i}}{4\sqrt{8}} \frac{1-i}{i}$$

Hence:

$$\begin{aligned}\mathrm{Re}\left(\int_{-\infty}^{\infty} R(x) e^{ix} dx\right) &= \mathrm{Re}(2\pi i(\mathrm{Res}_{z_1}(f) + \mathrm{Res}_{z_2}(f))) \\ &= 4\pi e^{-\sqrt{2}}\left(\cos\sqrt{2} - \sin\sqrt{2}\right)\end{aligned}$$

Integrals of rational function of trigonometric functions

Suppose $P(x,y), Q(x,y)$ are polynomials and Q is non-zero on the unit circle, i.e. $Q(x,y) \neq 0$ where $x^2 + y^2 = 1$ we want to compute:

$$\int_0^{2\pi} \frac{P(\cos t, \sin t)}{Q(\cos t, \sin t)} dt$$

We can re express this as a complex integral over the unit circle. Recall

$$\cos t = \frac{e^{it} + e^{-it}}{2}$$
$$\sin t = \frac{e^{it} - e^{-it}}{2i}$$

We have:

$$\begin{aligned}\int_0^{2\pi} \frac{P(\cos t, \sin t)}{Q(\cos t, \sin t)} dt &= \int_0^{2\pi} \frac{P\left(\frac{e^{it}+e^{-it}}{2}, \frac{e^{it}-e^{-it}}{2j}\right)}{Q\left(\frac{e^{it}+e^{-it}}{2}, \frac{e^{it}-e^{-it}}{2i}\right)} dt \\ &= \int_0^{2\pi} \frac{Q(\cdots)}{Q(\cdots)} \frac{1}{ie^{it}} ie^{it} dt\end{aligned}$$

Now we can set

$$f(z) = \frac{P\left(\frac{z+\frac{1}{z}}{2}, \frac{z-\frac{1}{z}}{2i}\right)}{Q\left(\frac{z+\frac{1}{z}}{2}, \frac{z-\frac{1}{z}}{2i}\right)} \frac{1}{iz}$$

Then the integral in question is equal to:

$$\int_0^{2\pi} f\left(e^{it}\right) ie^{it} dt = \int_\gamma f dz$$

Where $\gamma : [0, 2\pi] \to \mathbb{C} \quad \gamma(t) = e^{it}$

Since P, Q are polynomials,f is holomorphic on $C \setminus \{z_1, \ldots, z_k\}$ where $z_1, \ldots, z_k$ are the point where the denominator vanishes

Remark A function which is holomorphic except for isolated points is called **meromorphic**

If $z_1, \ldots, z_l$ are the points of $z_1, \ldots, z_k$ lying in the **interior of the unit circle**, the reside theorem implies:

$$\boxed{\int_0^{2\pi} \frac{P(\cos t, \sin t)}{Q(\cos t, \sin t)} dt = 2\pi i[\mathrm{Res}_{z_1}(f) + \cdots + \mathrm{Res}_{z_l}(f)]}$$

Example

Let us now try to compute $\int_0^{2\pi} \frac{1}{2\cos\theta} d\theta$. This falls into the previous example with $P(x, y) = 1$, $Q(x, y) = 2 + x$. Computing the function f from the previous example explicitly yields

$$f(z) = \frac{2}{i} \frac{1}{z^2 + 4z + 1}$$

The singular points are on the zeros of z^2+4z+1 which are $z_1 = -2+\sqrt{3}$ and $z_2 = -2-\sqrt{3}$. Only z_1 lies in the unit circle. Moreover f has a pole of order 1 at z_1. Hence

$$\mathrm{Res}_{z_1}(f) = \lim_{z \to z_1} (z - z_1) f(z) = \frac{1}{\sqrt{3}i}$$

Consequently,

$$\int_0^{2\pi} \frac{1}{2 + \cos\theta} d\theta = 2\pi i \mathrm{Res}_{z_1}(f) = \frac{2\pi}{\sqrt{3}}$$